

27. Differential equations:

Definition:

The equation which contains Variables derivatives

($\frac{d}{dx}$ form), not arbitrary constants is called as differential equation.

Example:

i. $\left(\frac{dy}{dx}\right)^3 - 5 \sin x \left(\frac{dy}{dx}\right)^2 = \left(\frac{dy}{dx}\right)^{10} + \cos y + x^2 = 0$

ii. $e^{\frac{dy}{dx}} = \sin(x+y)$

Order of differential equation:

(i). Order of DE is highest Order derivative which is Present in the given DE.

$\left(\frac{dy}{dx}\right)^6 - \left(\frac{dy}{dx}\right)^3 - \left(\frac{dy}{dx}\right) + 2 \sin x + 5y = 0$

: Order is 6.

(ii). $e^{y'''} = \sin(y+y') + y' - \cos x = 0 \Rightarrow$ Order 3.

Degree of DE:

: Degree of DE is Power of highest Order derivative in the given DE if DE is given in Polynomial form.

Otherwise, degree is not defined.

$0 = (x)^4 + \dots + \left(\frac{dy}{dx}\right)^4 + \left(\frac{dy}{dx}\right)^2 + \dots$

Ex: i. $\left(\frac{dy}{dx}\right)^5 - \left(\frac{dy}{dx}\right)^{10} + 5y\left(\frac{dy}{dx}\right)^3 - 7y = 0$

Order = 4

Degree = 3

Put $\frac{dy}{dx} = y_n$

$(y_n)^5 + (y_{n-2})^{10} + \dots = 0$

$(y_3)^5 + (-y_2)^{10} + 5y(y_4)^3 - 7y = 0$

Get into Standard Form...

ii. $\left(\frac{dy}{dx}\right)^4 - \sin\left(\frac{dy}{dx}\right) + y = 5$ Order = 3
Degree = N.D

iii. $\sin\left(\left(\frac{dy}{dx}\right)^2 + \left(\frac{dy}{dx}\right)^5\right) = x + 2$ Order = 3
Degree = ND

iv. $e^{\frac{dy}{dx^2}} = \left(\frac{dy}{dx}\right)^3 + y$ Order = 2
Degree = ND

Polynomial Form,

i. $a_n\left(\frac{dy}{dx}\right)^{k_1} + a_{n-1}\left(\frac{dy}{dx}\right)^{k_2} + \dots + F(x, y) = 0$
 $k_1, k_2, \dots, k_n \in \mathbb{Z}$

ii. Trigonometry Log $\rightarrow$ inside DE ND

28. Solution of DE :

1. Variable Separable Method :

let $\frac{dy}{dx} = f(xy)$, If $f(xy) = g(x) \cdot h(y)$

then $\frac{dy}{dx} = g(x) \cdot h(y)$ Separate 'x' terms to One Side

and 'y' terms to One Side and integrate to get the Solution...

Q. Solve the following DE :

i. $\frac{dy}{dx} = (1+x^2)(1+y^2)$

ii. $x^2 \frac{dy}{dx} = 2y+3$

iii. $\frac{dy}{dx} = e^{x-y} + x^2 e^{-y}$

i. $\frac{dy}{dx} = (1+x^2)(1+y^2)$

$\frac{dy}{1+y^2} = (1+x^2) dx$

Integrate both the sides...

$\int \frac{1}{1+y^2} dy = \int (1+x^2) dx$

$\tan^{-1} y = x + \frac{x^3}{3}$

$y = \tan\left(x + \frac{x^3}{3}\right)$

3. $\frac{dy}{dx} = e^{x-y} + x^2 e^{-y}$

$\frac{dy}{dx} = \frac{e^x}{e^y} + \frac{x^2}{e^y}$

$\frac{dy}{dx} = \frac{e^x + x^2}{e^y}$

$e^y dy = (e^x + x^2) dx$

Integrate ...

$\int e^y dy = \int (e^x + x^2) dx$

$e^y = e^x + \frac{x^3}{3}$

$\frac{x^3}{3} = e^y - e^x$

2.

$\frac{dy}{2y+3} = \frac{1}{x} dx$

2.

$$\frac{dy}{dx} = f(ax+by+c)$$

To solve this type of DE take $ax+by+c=t$
convert y in terms of t and x .

$$\frac{dy}{dx} = f(ax+by+c)$$

$$ax+by+c=t$$

$$a+b\frac{dy}{dx}+0=\frac{dt}{dx}$$

$$\frac{dy}{dx} = \frac{dt}{dx} - a$$

Ex: Solve $\frac{dy}{dx} = (x+y+2)^2$

Put $x+y+2=t$

$$1+\frac{dy}{dx}+0=\frac{dt}{dx}$$

$$\frac{dy}{dx} = \frac{dt}{dx} - 1$$

$$\int \frac{dt}{1+t^2} = \int dx$$

Integrate

$$\tan^{-1} t = x + C$$

$$\tan^{-1} (x+y+2) = x + C$$

Solve $\frac{dy}{dx} = x-y$

Homogeneous Differential Equation

let $f_1(x, y)$ and $f_2(x, y)$ are functions of x, y with same degree.

$$\boxed{\frac{dy}{dx} = \frac{f_1(x, y)}{f_2(x, y)}}$$

→ To Solve Above DE Take $\frac{y}{x} = v$ (or) $\frac{x}{y} = v$
and convert y in terms of v and x (or) x in terms of v and y ...

Q. Solve $\frac{dy}{dx} = \frac{2-y}{x}$

Equation reducible to Homogeneous diff eq :

$$\frac{dy}{dx} = \frac{ax+by+c}{Ax+By+D}$$

$$x \rightarrow x+h \quad y \rightarrow y+k$$

$$dx = dx$$

$$dy = dy$$

$$\frac{dy}{dx} = \frac{a(x+h)+b(y+k)+c}{A(x+h)+B(y+k)+D} = \frac{ax+by+ah+bk+c}{Ax+By+Ah+Bk+D}$$

Q:

5. Linear differential equation:

i. DE of form $\frac{dy}{dx} + P \cdot y = Q$

$$\frac{dy}{dx} + P \cdot y = Q$$

where P & Q are functions of 'x' only.

$$e^{\int P dx} \left(\frac{dy}{dx} + P \cdot y \right) = e^{\int P dx} \cdot Q$$

$$e^{\int P dx} \cdot \frac{dy}{dx} + e^{\int P dx} \cdot P \cdot y = e^{\int P dx} \cdot Q$$

$$\frac{d}{dx} (y \cdot e^{\int P dx}) = Q \cdot e^{\int P dx}$$

$$\int \frac{d}{dx} (y \cdot e^{\int P dx}) = \int Q \cdot e^{\int P dx} dx$$

$$y \cdot e^{\int P dx} = \int Q \cdot e^{\int P dx} dx$$

Solution

of
given DE

$$y \cdot (I.F) = \int Q \cdot (I.F) dx$$

Q: Solve $x \log x \frac{dy}{dx} + y = 2 \log x$

$$\frac{dy}{dx} + \frac{1}{x \log x} \cdot y = \frac{2}{x}$$

$$P = \frac{1}{x \log x}$$

$$Q = \frac{2}{x}$$

$$e^{\int P dx} = e^{\int \frac{1}{x \log x} dx} = e^{\log t} = t = \log x$$

$$y \cdot \log x = \int \frac{2}{x} \log x dx$$

$$y \cdot \log x = 2 \int t dt = \frac{2}{2} t^2 + C = t^2 + C = (\log x)^2 + C$$

ii) Bernoulli's equation

$$\frac{dy}{dx} + Py = Qy^n$$

$$\frac{1}{y^n} \cdot \frac{dy}{dx} + P \cdot y^{1-n} = Q$$

$$\frac{1}{y^n} \frac{dy}{dx} + P \left(\frac{1}{y^{n-1}} \right) = Q$$

$$\text{Put } \left(\frac{1}{y^{n-1}} \right) = t$$

$$\frac{1}{1-n} \frac{dt}{dx} + Pt = Q$$

$$y^{1-n} = t$$

$$(1-n)(y)^{-n} \frac{dy}{dt} = \frac{dt}{dx}$$

$$\boxed{\frac{dt}{dx} + P(1-n)t = Q(1-n)}$$

$$\frac{2 \sin 4x}{\sin^2 2x} \quad \frac{2 \sin 4x}{\sin 2x \sin 2x}$$

$$\frac{2 \cdot 2 \cancel{\sin 2x} \cos 2x}{\cancel{\sin 2x} \cdot \sin 2x}$$

$$4 \cot 2x$$